EXP3:NETWORK TOPOLOGY CREATION NS3

RAVI BALA M.V
Register Number: 24BCE1419

Faculty Name: Subbulakshmi

Lab Slot: L55 + L56

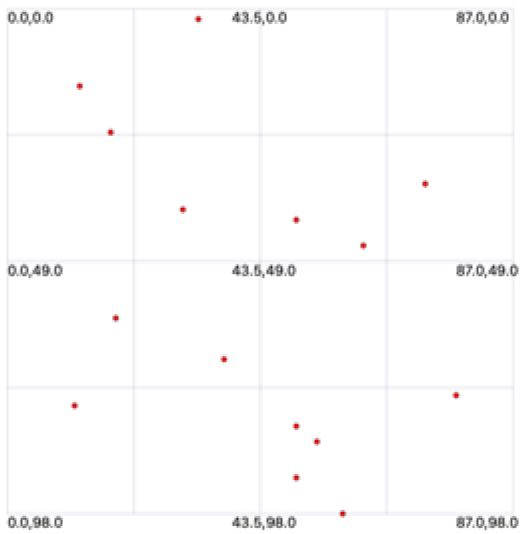
Course Name: Computer Networks Lab
Course Code: BCSE308L

BUS TOPOLOGY:

```
#include "ns3/core-module.h"

#include "ns3/network-module.h"

#include "ns3/csma-module.h"

#include "ns3/internet-module.h"

#include "ns3/netanim-module.h"

using namespace ns3;

int main() {

     NodeContainer nodes;

     nodes.Create(15);

     CsmaHelper csma;

    csma.SetChannelAttribute("DataRate", StringValue("100Mbps"));

    csma.SetChannelAttribute("Delay", TimeValue(NanoSeconds(6560)));

     NetDeviceContainer devices = csma.Install(nodes);

     InternetStackHelper internet;

     internet.Install(nodes);

     Ipv4AddressHelper ipv4;

    ipv4.SetBase("10.1.1.0", "255.255.255.0");

     ipv4.Assign(devices);

     AnimationInterface anim("bus.xml");

     Simulator::Run();

Simulator::Destroy();

     return 0;
```

```
}
```

O/P:

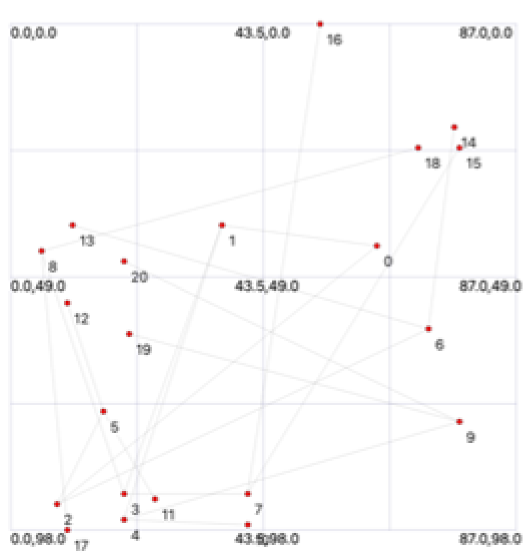


TREE TOPOLOGY:

```
#include "ns3/core-module.h"

#include "ns3/network-module.h"

#include "ns3/point-to-point-module.h"

#include "ns3/internet-module.h"

#include "ns3/netanim-module.h"

using namespace ns3;

int main ()

{

      NodeContainer nodes;

      nodes.Create (21);

      InternetStackHelper internet;
```

```
    internet.Install (nodes);
    PointToPointHelper p2p;
  p2p.SetDeviceAttribute ("DataRate", StringValue ("2Mbps"));
  p2p.SetChannelAttribute ("Delay", StringValue ("2ms"));
    for (uint32_t i = 1; i < 21; i++)
    {
    p2p.Install (nodes.Get ((i - 1) / 2), nodes.Get (i));
    }
    AnimationInterface anim ("tree.xml");
    Simulator::Run ();
    Simulator::Destroy ();
    return 0;
}
```

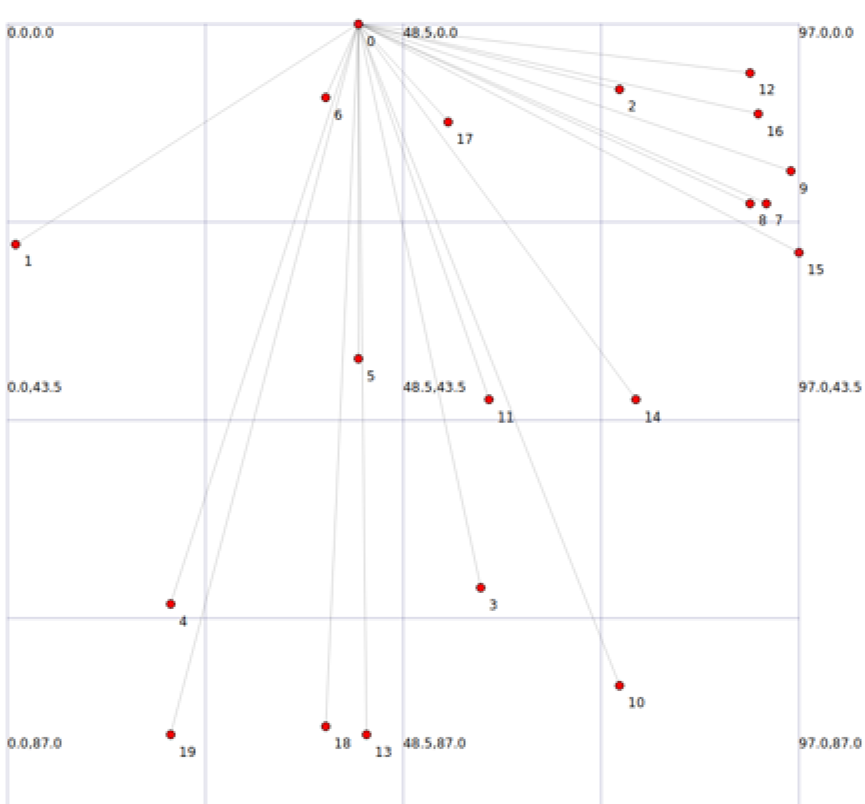
O/P:



```
#include "ns3/core-module.h"

#include "ns3/network-module.h"

#include "ns3/point-to-point-module.h"

#include "ns3/internet-module.h"

#include "ns3/netanim-module.h"

using namespace ns3;

int main() {

    NodeContainer hub;

    hub.Create(1);

    NodeContainer spokes;

    spokes.Create(19); // Total = 20 nodes

    InternetStackHelper stack;

    stack.Install(hub);

    stack.Install(spokes);

    PointToPointHelper p2p;

    p2p.SetDeviceAttribute("DataRate", StringValue("5Mbps"));

    p2p.SetChannelAttribute("Delay", StringValue("2ms"));

    Ipv4AddressHelper address;

    for (uint32_t i = 0; i < spokes.GetN(); i++) {

      NodeContainer pair(hub.Get(0), spokes.Get(i));

      NetDeviceContainer devices = p2p.Install(pair);

      std::ostringstream subnet;

      subnet << "10.1." << i+1 << ".0";
```

```
      address.SetBase(subnet.str().c_str(), "255.255.255.0");
      address.Assign(devices);
    }
    AnimationInterface anim("star.xml");
    anim.SetConstantPosition(hub.Get(0), 50, 50);
     for (uint32_t i = 0; i < spokes.GetN(); i++) {
      anim.SetConstantPosition(spokes.Get(i), 5*i, 10);
    }
    Simulator::Run();
    Simulator::Destroy();
}
```

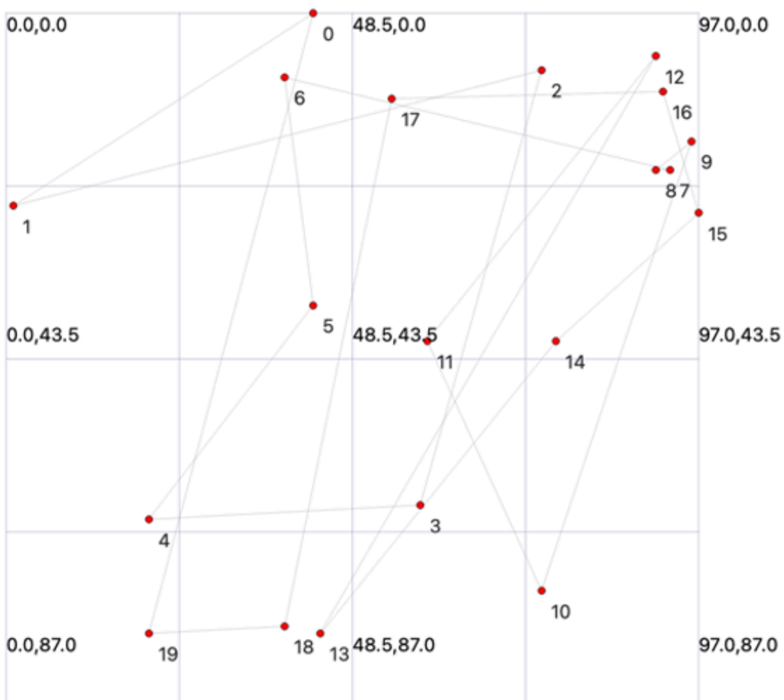
O/P:



```
#include "ns3/core-module.h"

#include "ns3/network-module.h"

#include "ns3/point-to-point-module.h"

#include "ns3/internet-module.h"

#include "ns3/netanim-module.h"

using namespace ns3;

int main ()

{

    NodeContainer nodes;

    nodes.Create (20);

    InternetStackHelper internet;

    internet.Install (nodes);

    PointToPointHelper p2p;

    p2p.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));

    p2p.SetChannelAttribute ("Delay", StringValue ("2ms"));

    for (uint32_t i = 0; i < 19; i++)

    {

      p2p.Install (nodes.Get (i), nodes.Get (i + 1));

    }

    p2p.Install (nodes.Get (19), nodes.Get (0));

    AnimationInterface anim ("ring.xml");

    Simulator::Run ();

    Simulator::Destroy ();
```

```
    return 0;

}
```

O/P:

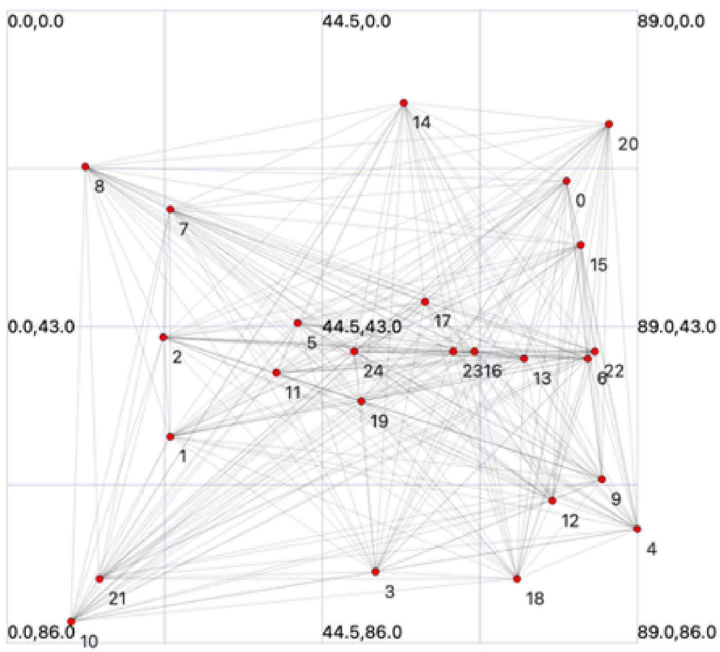


MESH TOPOLOGY:

```
#include "ns3/core-module.h"

#include "ns3/network-module.h"

#include "ns3/point-to-point-module.h"

#include "ns3/internet-module.h"

#include "ns3/netanim-module.h"

using namespace ns3;

int main ()

{
```

```
NodeContainer nodes;
nodes.Create (25);
InternetStackHelper internet;
internet.Install (nodes);
PointToPointHelper p2p;
p2p.SetDeviceAttribute ("DataRate", StringValue ("1Mbps"));
p2p.SetChannelAttribute ("Delay", StringValue ("3ms"));
for (uint32_t i = 0; i < 25; i++)
{
for (uint32_t j = i + 1; j < 25; j++)
{
    p2p.Install (nodes.Get (i), nodes.Get (j));
}
}
AnimationInterface anim ("mesh.xml");
Simulator::Run ();
Simulator::Destroy ();
return 0;
}
```

O/P:



0.0,0.0
44.5,0.0
89.0,0.0
0.0,43.0
44.5,43.0
89.0,43.0
0.0,86.0
44.5,86.0
89.0,86.0
14
20
8
0
7
15
17
5
2
24
23
16
13
6
22
11
19
1
9
12
4
21
3
18
10